

A building regulation question answering system: A deep learning methodology

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ABSTRACT

Regulations play an important role in assuring the quality of a building's construction and minimizing its adverse environmental impacts. Engineers and the like need to retrieve regulatory information to ensure a building conforms to specified standards. Despite the availability of search engines and digital databases that can be used to store regulations, engineers, for example, are unable to retrieve information for domain-specific needs in a timely manner. As a consequence, users often have to deal with the burden of browsing and filtering information, which can be a time-consuming process. This research develops a robust end-to-end methodology to improve the efficiency and effectiveness of retrieving queries pertaining to building regulations. The developed methodology integrates information retrieval with a deep learning model of Natural Language Processing (NLP) to provide precise and rapid answers to user's questions from a collection of building regulations. The methodology is evaluated and a prototype system to retrieve queries is developed. The paper's contribution is therefore twofold as it develops a: (1) methodology that combines NLP and deep learning to be able to address queries raised about the building regulations; and (2) chatbot of question answering system, which we refer to as QAS4CQAR. Our proposed methodology has powerful feature representation and learning capability and therefore can potentially be adopted to building regulations in other jurisdictions.

1. Introduction

Regulations play an important role in assuring a building's quality [1], but they are often difficult to retrieve as they organized in a text format accordance with a rigid classification system [2,3]. Tools, such as question answering (QA) systems, are therefore needed to respond to a user's questions. Knowledge-Based Question Answering (KBQA) systems, for example, have been developed to transform a question from its natural language into an internal representation to capture its semantics and intent by using knowledge representation templates. Usually a deep relevant of user questions and answer passages is needed to extract the relevant information and place into templates [4].

Despite the use of search engines for raising queries, users have to

deal with the burden of having to browse and filter out results of find the information that they require [5]. While these search engines may well improve accessibility to regulations, candidate passages that are identified by the process of retrieval can often be verbose and not necessarily of value to person (e.g., engineer) that has raised the query. The task of question answering aims to provide users with precise answers to questions raised in a timely manner. In this paper, a robust end-to-end methodology to improve the efficiency and effectiveness of retrieving building regulation queries is developed.

The paper commences by providing a review of the related research on the modeling and retrieval, extraction, and question answering of regulations. Then, a methodology for answering building regulation queries that combines the retrieval of information and deep learning

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Natural Language Processing (NLP) is proposed. The developed methodology can be used to facilitate the extraction of precise answers from a clause (passage, corpus) of regulation text by focusing: (1) factoid; and (2) definition, which are a form a proportion of the clauses in 19 regulations (i.e., approximately 90%). The methodology is subsequently evaluated using an experiment and a prototype system (chatbot) is proposed.

2. Related research

Building regulations set standards for the design and construction of buildings to ensure the safety and health for people. In a nutshell, regulations set national standards for building work, whether it will be on a major new development or an extension or alterations. They cover all aspects of construction, including foundations, damp-proofing, the overall stability of the building, insulation, ventilation, heating, fire protection and means of escape in case of fire. Regulations are complex and therefore being able to facilitate the retrieval of relevant information in an expeditious manner can aid the process of assuring quality.

2.1. Regulation modeling and retrieval

Research examining regulation knowledge modeling and retrieval in buildings has received widespread attention. Building regulations are typically presented in a text format (i.e., paper or digital), rendering it difficult for a computer to process. Thus, to provide additional computerized support to the users of regulations, representing regulation knowledge in a formal and computer-interpretable way is necessary [3]. A variety of modeling approaches have been applied to represent the logic of building regulations [6], which include object-oriented [7], rule-based [8], ontology-based [9] and semantic-based [10].

A detailed review of regulation knowledge modeling and automatic rule-based checking can be found in the works of Zhong *et al.* [3], Eastman *et al.* [8] and Pauwels *et al.* [10]. It is difficult to represent all the regulations using a singular approach. Regulations, for example, can be represented by both qualitative (e.g., ‘the anchorage body should adopt cement paste or mortar’) and quantitative (e.g., quantity value and unit) constraints [11]. In the case of qualitative constraints, a variety of syntaxes can be used for the same purpose. Additionally, most modeling and representation methods in formalization (including the extraction of logic and rules from regulatory texts) are conducted manually or semi-automatically, which is time-consuming and dependent on manual feature engineering.

Numerous approaches have been put in place to help retrieve regulations, which include the creation of retrieval taxonomies [12], use of controlled vocabularies [13], and development of ontologies [14]. However, directly applying retrieval techniques often results in a list of relevant documents being produced, which have to be sorted accordingly to obtain the information required [14].

2.2. Information extraction from regulations

Natural language processing enables a computer to understand and process natural language text in a human-like manner. Information extraction (IE) is a subfield of NLP that deals with extracting desired information from text sources. Information extraction can generally be achieved in two ways [11]: (1) machine learning (ML); and (2) rule-based approaches. However, in light of the sparsity of training samples, research has tended to focus on the use of rule-based approaches. Such approaches require less training data than statistical methods and enable users to gain additional insights into certain relations [15].

There has been limited research that has examined the IE of clauses from building regulations. The development of extraction rules is time consuming and manual laden process. Moreover, extracted rules are difficult to be reuse as they are domain-dependent and tend to be defined by specific regulatory documents. Ontology-based information

extraction (OBIE) can be used to address the shortcomings of IE as it uses the domain and application ontology to assist in extracting semantic information that is specific to a particular domain and application [16,17]. Zhang and El-Gohary [18,19], for example, utilized semantic modeling and NLP techniques to facilitate the automation the processing and analysis of regulation documents analysis (e.g., regulation analysis) and processing. This then enabled Zhang and El-Gohary [18,19] to extract information requirements from documents and formalize them into a computer-processable format.

Most IE based studies in construction have focused on the use of simple quantitative constraints, or describing explicit patterns or templates [11]. There has however been limited research that has used ML-based regulation extraction as this approach is data-intensive and requires learning samples for training an effective extraction model. From a user's perspective, a large amount of data is not always readily available, while (general) domain knowledge is usually at hand.

A methodology for extracting information entities (e.g., data on bridge conditions and maintenance actions) from bridge inspection reports has been developed based on an ontology and semi-supervised conditional random fields (CRF) [20]. However ML-based algorithms (e.g., support vector machines (SVM), hidden Markov model, or CRF) are feature-based and thus need complicated feature engineering to select effective handcrafted features to obtain improved IE performance [21].

Deep neural network-based methods are able to automate the process of feature learning. In comparison with handcrafted feature-based approaches, deep learning methods can learn related features from given sentences without undertaking complicated feature engineering work [22]. The use of deep learning for extracting building regulation information extraction is in its infancy [11].

Contrastingly, contextualized embeddings produce word vector representations that dynamically change with the appearance of words via the use of transfer learning mechanisms [23]. Contextualized embeddings have been used to obtain state-of-the-art results for a wide variety of NLP tasks [24,25]. However, to the authors knowledge there have been no studies that have specifically analyzed building documentation (such as regulations) or utilized modern NLP techniques (such as transformers, including bidirectional encoder representations from transformers). Against this backdrop, this research explores how a hybrid deep neural network, which combines the attention mechanism and the contextualized representation, can be used to extract words within the text considering their context.

2.3. Question answering

A common approach for question analysis is to map them into a predefined template (e.g., ‘What-type’ and ‘How-type’) [26]. Existing online question answering search engines utilize this approach. Nonetheless, template-based approaches have significant limitations as they are unable to handle multiple questions. In contrast to the traditional template-based question answering methodology, which focuses on the specific knowledge presentation patterns of text document and has a small corpus with low scalability, deep learning-based question answering (e.g. convolutional neural networks (CNN)), can represent and analyze questions automatically. The neural model is normally trained via an end-to-end approach to obtain an answer to the given question and passage.

Deep learning can extract and analyze complex characteristics presented in textual documents and learn representations from input data [27]. An innate feature of deep learning-based text classification, for example, is the ability to accurately extract word order information and then further learn from the semantic information contained within the text. A case in point can be found in the work undertaken by Yu and Karl [28] who utilized a CNN to model the distribution of questions and answers. Similarly, Bahdanau *et al.* [29] introduced an attention mechanism into the question answering task due to its superior

performance in a sequence-to-sequence model. Needless to say, deep learning approaches require a large amount of training data [30], which have been unavailable for building regulation question answering tasks.

Attention (i.e., these mechanisms orient perception and memory access within a neural network) and transfer learning mechanisms have been introduced and developed to further increase the representation flexibility of deep learning models [31]. Many successful deep neural network models have significantly benefitted from transfer learning and attention mechanisms on the topic of text classification [32] and named entity recognition [33]. The recent development of Bidirectional Encoder Representations from Transformers (BERT) [34] has proven the effectiveness of contextualized representations from a deep structure (e.g., bidirectional language model) for transfer learning. The BERT extracts contextual meaning through bidirectional encoding with a masked language model and a next-sentence prediction objective. Day *et al.* for example, proposed a system architecture of factoid question answering system using BERT with question expected answer type and answer type classification models [35]. Similarly, Bolotova *et al.* fine-tuned a BERT model on a non-factoid question answering task to determine if a model was able to attend to words similar to those annotated [36]. As an emerging model, BERT-based models have been used for medical domain question answering tasks achieving the state-of-the-art performance without any handcrafted features [37].

Studies examining question answering for building regulations are scarce [38]. As a consequence, a methodology for building regulation question answering is proposed and developed in the next section of this paper.

3. Methodology

An overview of the proposed methodology for question answering of the building regulation is shown in Fig. 1. The core architecture comprises the following two main modules: (1) the IR and (2) a pretrained BERT. The IR module is responsible for selecting segments of text that contain answers. These segments are combined and then passed to the pretrained BERT module to identify an answer span in the sentence clause. The main steps of the methodology are as follows: (1) data pre-processing, (2) dataset creation, (3) passage retrieval, and (4) BERT module training and testing.

The first step provides a corpus for the QA system, and removes meaningless words and converts questions and passages into suitable data structures for further computation and processing, which occurs in the third step. The second step develops a triple dataset to train. The third step is to search for some passages related to the question and output the top-ranked passages. In fourth step, the BERT model is pre-trained to analyze the natural language input of users and generate appropriate answers. Finally, the methodology and a prototype are implemented and evaluated.

3.1. Data Pre-processing

The Chinese building regulations were downloaded in a portable document format. Notably each clause is treated as a passage (hereafter clause and passage are used indiscriminately). The clauses were extracted manually from the regulations. Then, each clause was represented in a separate text format file and a triple set comprising passage, question, and answer.

Most NLP approaches focus on mining texts from varying types of documents, but the semantic information that is embedded in the tables and lists produced is often lost. Table 1 presents a snapshot of the regulation form for inspecting the quality of a concrete structure. The cells (e.g., “item,” “allowable deviation,” and “5 mm”) in the table have minimal meaning when viewed separately, where the + 10, 0 means that the allowable deviation can only be positive 0 to 10 and not negative. However, when it is considered as a whole, the content of the table can help correctly answer questions, such as “What is the allowable error for the position of the centerline of the embedded steel pipe?” The pre-processing approach developed by Cao *et al.* [26] was adopted to retain semantic information extracted from tables and lists.

For each table, all the textual information in each row or column (depending on the header location), together with the corresponding header and caption text, are formed as a separate passage to be indexed by the system. Each list is treated as a tree, and each non-leaf branch node (including the root node) generates a separate passage to be indexed by collapsing all contained nodes. Next the corresponding textual information is combined with the caption text. Additionally, titles may be merged with any clause within their corresponding section to capture the hierarchical nature of clauses in regulations and to retain semantic information based on the assumption that they carry important semantic content. Though, this content may not necessarily be explicitly described in narrative passages.

Before a question is raised and the corresponding passage retrieved, preprocessing through word segmentation and stop word removal are required. The preprocessing procedure is implemented in Jieba (a

Table 1

Regulation form for quality inspection of concrete structure.

Item	Allowable Deviation (mm)
Centerline position of embedded steel plate	3
Centerline position of embedded pipe and reserve hole	3
Steel dowel	Centerline position
	Leakage length
Embedded bolt	Centerline position
	Leakage length
Reserved hole	Centerline position
	Size

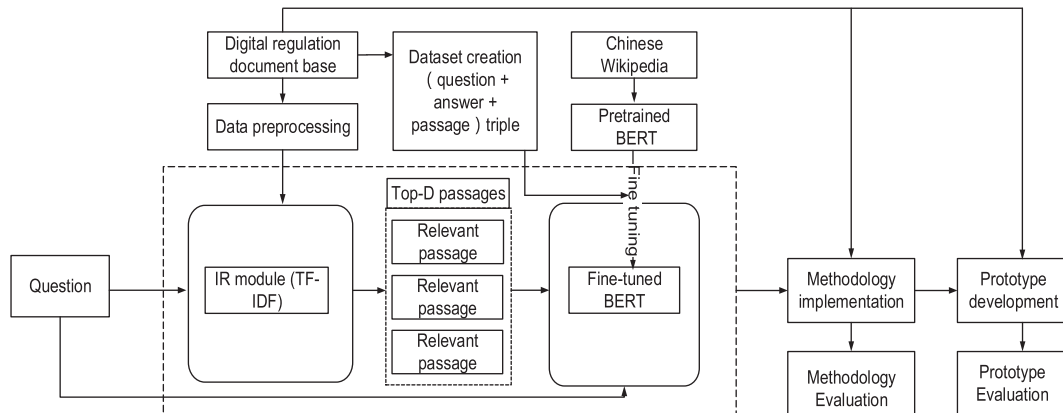


Fig. 1. Overview of the methodology.

Chinese word segmentation software based on Python). A typical feature of the Chinese script is that there is no blank space between characters in a sentence, which results in word segmentation producing differing results and interpretations of the text.

For example, “钢筋混凝土 (reinforced concrete)” could be interpreted as “钢筋 (bar)” and “混凝土 (concrete).” Inaccurate word segmentation can lead to ambiguities and thus hinder the interpretation of the information. To overcome this, a function provided by Jieba enables users to create their own dictionaries. In this research, the dictionaries, such as a “safe production vocabulary .scel” created by the Infrastructure Construction Department of HuBei Province, China, and the “architectural vocabulary .scel” generated and customized from Chinese National Building Standards, are utilized. Some words have special semantics and therefore are segmented and retained due to the language syntactic features of the regulations. Here the following word taxonomies are listed as examples.

1. Deontic taxonomy comprises must(必须), must not(严禁), shall(应), shall not(不应), should(宜), should not(不宜), may(可), and need not(不必), which indicate various levels of restrictions.
2. Comparative taxonomy includes greater/bigger than (大于), smaller than (小于), not greater/bigger than (不大于), at most (至多), at least (至少), lower than (低于), not lower than (不低于), less than (少于), and not less than (不少于).
3. Temporal taxonomy comprises before (前), after (后), former (之前), latter (之后), when (当...时候), immediately (立即), and right away (马上), which are used in the procedure constraints to control the temporal relationships between construction activities. In addition, some verbs, such as construct (施工), complete (完成), realize (实现), and start (开始), are usually used in conjunction with the temporal terms, such as implementing (在进行) before (前)/after (后), starting (在开始) before (前)/after (后), and finishing (在完成) before (前)/after (后).

With the assistance of dictionaries, words can be segmented accurately with a “word group”, so that sentences can be interpreted correctly. Furthermore, the “HIT stopwords list .txt” and the “MIL-SCU stopwords list .txt” are commonly used to remove stopwords in Chinese. The both lists have many common stopwords and also some different stopwords, the combination of the both lists is to cover almost all the Chinese stopwords to improve the performance of pre-processing. By building a stopwords dictionary, words such as “an”, “some”, are removed. In doing so, only valid words of text remain to reveal its distinctive features[39].

3.2. Passage retrieval

Each clause/passage of regulation is treated as a “document” (i.e., the unit of retrieval), and the regulation documents are pre-segmented into a corpus of passages. The passage retrieval function selects passages related to the question.

The IR module lies at the heart of the passage retrieval process and is used to identify relevant passages from the building regulation corpus. Furthermore, passages are ranked to identify relevant text segments and then extracted to answer the raised question. Different models, including Boolean model [40], vector space model (VSM) [41], ontology-based approach [42], latent semantic indexing [43], and language model [44], have been developed for information retrieval.

The information retrieval model developed by Jayashree *et al.* [45], which utilized a VSM [41] is akin approaches used for passage retrieval. Here the VSM model is used to convert each passage and question into a vector, which is formed by calculating the Term Frequency-Inverse Document Frequency (TF-IDF) of terms in each document. The TF-IDF provides each keyword with a weight based on the following two criteria: (1) the frequency of its usage in the specified document (TF); and (2) the rarity of its appearance in the other documents in the corpus

(IDF) [46]. Then, with the cosine similarity between the question and the document calculated, passages related to the question are arranged as candidates according to the size of similarity.

3.3. BERT module training and testing

Questions and passages must be inputted into the BERT module to extract the proper answer from the top D passages of the aforementioned output. The input of questions and passages must first satisfy the form of the BERT model, which is “[CLS] question [SEP] context [SEP].” In this model, [CLS] is a special symbol added in the front of every input example and [SEP] is a special separator token (e.g., separating questions/answers). Each retrieved passage and question are packed and converted into the BERT for embedding. The architecture of fine-tuning BERT for building regulation QA is presented in Fig. 2.

The BERT-based model architecture is a multi-layer bidirectional transformer structure comprising the following three layers: (1) word embedding; (2) encoding; and (3) answer layers. The word embedding layer, which comprises position, segment, and token embeddings, is designed to turn the question and passage into sequences of words. The position embeddings are used to provide location information for transformers in the encoding layer. The segment embeddings distinguish between the question and passage. And, the token embeddings are employed to map each word into a high-dimensional vector space to fulfill the training requirement of BERT.

The encoding layer consists of six identical layers. Each layer has two sub-layers: (1) a multi-head self-attention mechanism; and (2) a fully connected feed-forward network with simple and position-wise. The multi-head attention sublayer plays an important role in the transformer’s architecture. It utilizes an attention mechanism, which relates different positions of a single text to it and then computes a better representation. The attention mechanism [31] is widely used in NLP, which maps a query vector Q and a set of key-value vector pairs (K, V) to an output, described as follows:

$$O_a = \text{Attention}(Q, K, V) \quad (1)$$

The Scaled Dot-Product Attention is used in multi-head attention to achieve the question-aware passage embedding so as to catch inner-sequence information, which is formulated as:

$$O_a = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

where d_k denotes the dimension of keys K and values V. As in Eq. (2), the Scaled Dot-Product Attention utilizes a Scaled Dot-Product with division d_k to calculate similarity, which is the difference between the general attention model for adjustment that prevents the inner product from becoming too big. The multi-head attention employs n different linear layers to linearly map vector Q, K, V and then focuses on obtaining multi-head attention outputs, which is expressed as:

$$O_m(Q, K, V) = \text{concat}(h_1, h_2, \dots, h_n)W^o \quad (3)$$

$$h_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), i = 1, \dots, n \quad (4)$$

where W^O, W^K, W^Q and W^V denote the linear layer weights for output, vectors Q, K and V respectively. We use the same self-attention mechanism (or intra-attention) and vector Q, K and V in our work.

The fully connected layer possesses a position-wise mechanism. Since this layer processes the attention position i in the Multi-head Self-attention, the feed-forward network is referred to as being ‘Position-wise’. The position-wise feed-forward networks have two linear layers with a Rectified Linear Unit (ReLU) layer between them, which is summarized as:

$$O_P = [\max(xW_1^P + b_1^P, 0)]W_2^P + b_2^P \quad (5)$$

In the encoder, the weights $W_i^Q \in \mathbb{R}^{d_m \times d_k}$, $W_i^K \in \mathbb{R}^{d_m \times d_k}$, $W_i^V \in \mathbb{R}^{d_m \times d_v}$,

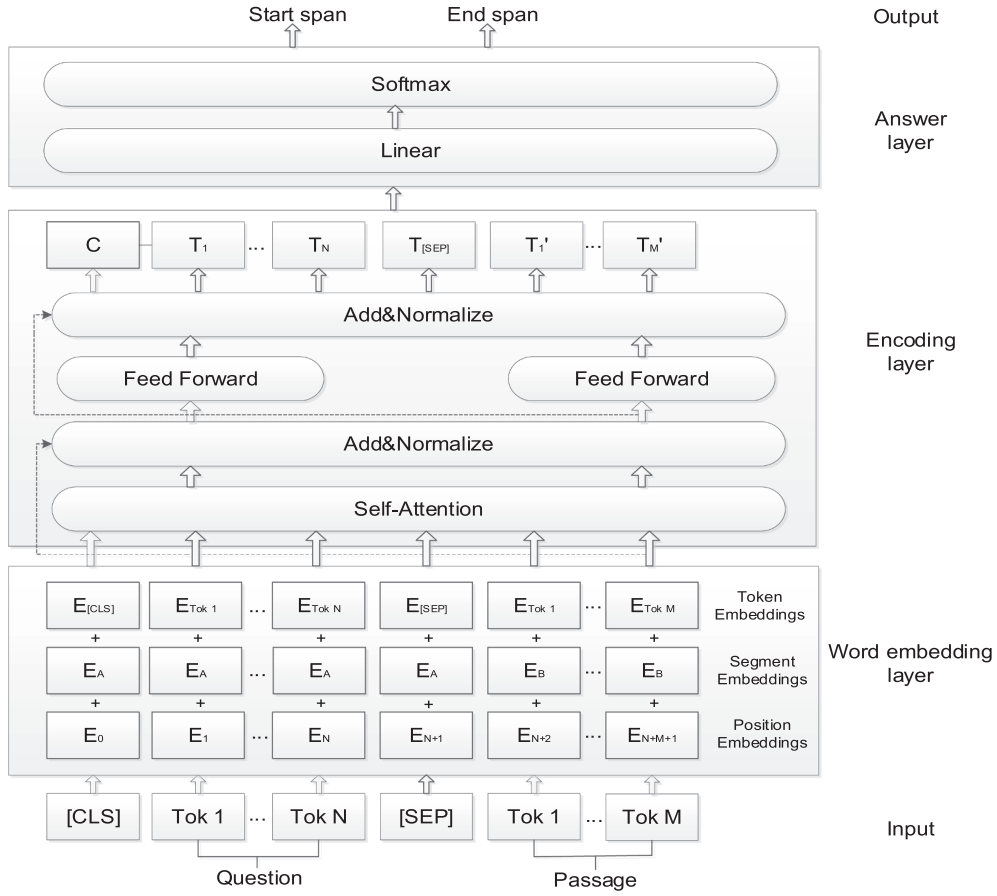


Fig. 2. Architecture of fine-tuning BERT for building regulation QA.

and $W_i^o \in \mathbb{R}^{nd_v \times d_m}$, where $d_m = 768$ in this work denotes the dimension of input data.

The answer layer is to predict the answer span with the start and the end positions by the highest probability, which is consist of Linear and SoftMax. The Linear and SoftMax take contextual word embeddings from BERT as input. Specifically, the probabilities of the t -th token being the start of the answer token P_t^s can be calculated by the following equation:

$$P_t^s = \frac{e^{V_s \cdot T_t}}{\sum_j e^{V_s \cdot T_j}} \quad (6)$$

where $T_l \in \mathbb{R}^H$ denotes l -th token representation from BERT and “ $\cdot$ ” denotes the dot product between two vectors, the trainable start vector as $V_s \in \mathbb{R}^H$ where H denotes the hidden size of BERT. The probability of a token being the end token is computed likewise. Further details regarding the structure and nature of the transformer model can be found in Devlin *et al.* [34] and Vaswami *et al.* [47].

Model pre-training plays an important role in improving the accuracy and reliability of the developed model. The performance of language representation models largely depends on the size and quality of corpora on which they are pre-trained. Given that BERT is designed as a general-purpose language representation model, its base model (uncased, 12 layers, 768 hidden, 12 heads, 110 M parameters) is first used to apply this model to Chinese texts. Then it is pre-trained using the Chinese Wikipedia. All inputs are padded to 256 tokens, the learning rate is set to 3×10^{-5} , and all other default settings are used. The length of most passages (up to 98%) obtained during the data preparation is less than 256. Considering the computing resource limitation, the token length is set to 256. In this case, if the packed length of the pair of passage and question is longer than 256, then the extra length of the

sentence will be automatically deleted. Finally, the pretrained BERT model by Chinese Wikipedia is further trained on the dataset created regulation texts to generate the QA model. This fine-tuning process is discussed in Section 4 below.

4. Implementation of the proposed methodology

The implementation of the proposed QA methodology involves data collection, dataset creation and model training and testing.

4.1. Data collection

Chinese building regulations often intersect with one another, which can be grouped into specific categories. The detailed statistical results are shown in Table 2. Considering the pertinence and practicability of the research, 19 regulations are selected as the object of this research, as

Table 2
Classification statistics of building regulations.

Regulation type	Number of regulations	Regulation type	Number of regulations
Principal structure class	25	Materials and application class	24
Quality inspection class	19	Safety and health class	19
Decoration class	17	Professional engineering class	13
Foundation and base classes	13	Detection technology class	13
Technology class	12	Construction organization and management class	8

listed in Table 3.

4.2. Dataset creation

To the authors knowledge there is no standard and manually-annotated corpus is available for training and evaluating this type of regulation question answering task.

The domain-specific dataset of the building regulations for model training is created as follows. While the proposed approach can solve different types of questions (without complex manual feature engineering for different questions and answers), analyzing these questions is helpful in creating the training dataset. Some studies have analyzed different types of questions and answers in their specific research fields or focused domain. For example, Hyo-Jung and Ki-Youn [48] identified six surface-level question types: 5W1H, that is, who, what/which, where, why, when, and how. These types are determined on the basis of the specific domain and purpose.

Most regulation clauses consider definitions and descriptions of techniques or compliance checking requirements. Two types of clause standout:

1. Definition: This type of clause is a definition of a concept, which is recognized using a set of simple rules on user questions, such as “What is X? What does X mean?” (where X is a noun phrase). In this case, X is the focus of the question. The simplified form of the question will be X. For example, what is the master project? The key word in this question is master project. The answer is often the entire clause in the regulation, which usually longer than that of the factoid answer.

Table 3

Source of data collection: 19 regulation documents.

No	Chinese Building Regulations
1	GB50300 《建筑工程施工质量验收统一标准 (Unified standard for quality inspection of building engineering)
2	GB50202 《建筑地基基础工程施工质量验收标准 (Standard for inspection of quality of building foundation)
3	GB50203 《砌体结构工程施工质量验收规范 (Regulation for inspection of quality of masonry structures)
4	GB50204 《混凝土结构工程施工质量验收规范 (Regulation for quality inspection of concrete structure)
5	GB50205 《钢结构工程施工质量验收规范 (Regulation for inspection of quality of steel structures)
6	GB50206 《木结构工程施工质量验收规范 (Regulation for inspection of quality of timber structures)
7	GB50207 《屋面工程质量验收规范 (Regulation for inspection of quality of roof)
8	GB50208 《地下防水工程质量验收规范 (Regulation for inspection of quality of underground waterproof)
9	GB50209 《建筑地面工程施工质量验收规范 (Regulation for inspection of quality of building ground)
10	GB50210 《建筑装饰装修工程质量验收标准 (Standard for inspection of quality of building decoration)
11	GB50242 《建筑给水排水及采暖工程施工质量验收规范 (Regulation for inspection of quality of water supply drainage and heating works)
12	GB50243 《通风与空调工程施工质量验收规范 (Regulation for inspection of quality of ventilation and air conditioning works)
13	GB50303 《建筑电气工程施工质量验收规范 (Regulation for inspection of quality of building electrical engineering)
14	GB50310 《电梯工程施工质量验收规范 (Regulation for inspection of installation quality of lifts, escalators, and passenger conveyors)
15	GB/T50375 《建筑工程质量评价标准 (Evaluation standard for excellent quality of building engineering)
16	GB50339 《智能建筑工程质量验收规范 (Regulation for quality inspection of intelligent building system)
17	GB50411 《建筑节能工程施工质量验收规范 (Regulation for inspection of energy-efficient building)
18	GB50312 《综合布线系统工程验收规范 (Regulation for engineering inspection of generic cabling system)
19	JGJ/T139 《玻璃幕墙工程质量检验标准 (Standard for testing of engineering quality of glass curtain walls)

2. Factoid: This type of clause is description of techniques or compliance checking requirements, which is recognized using a set of simple rules on user questions (e.g., firstWordOf (Q) = (How/What/When/Where/Who/Why) indicates that question Q is a 5W1H question) and patterns. This clause determines the expected answer type and the way of the triple construction. For example, what is the function of mortar when building blocks? The answer is often a specification entity and information, which is some fragment from a clause in the regulation that always belongs to short texts [49].

A strategy often used to create the dataset is to ask as many questions as possible for a given answer. In this instance the trained model is flexible and capable of solving different questions. For example, for the answer (clause) “伸缝:为防止水泥混凝土垫层在气温升高时在缩缝边缘□生挤碎或拱起而设置的伸胀缝” (Extension joints are set up to prevent cement concrete cushion from crushing or arching at the edge of contraction joints with increased temperature), the following three different styles of questions can be given:

1. What is an extension joint?
2. What is the function of extension joints?
3. Can the extension joints prevent the cement concrete cushion from crushing or arching at the edge of the contraction joints when the temperature rises?

For another answer (clause) “In the construction of underground waterproofing works, a process of self-inspection, handover inspection, and full-time personnel inspection shall be established, and there shall be a complete inspection record”, two corresponding different styles of questions with the same answer are presented below:

1. How should a contractor carry out the process of underground waterproofing works?
2. What is the process of underground waterproofing works ?

The building regulations is used to train and evaluate the proposed system and construct the triples dataset of the passage-question-answer. The triples dataset is constructed in accord with Stanford Question Answering Dataset (SQuAD) [50]. The SQuAD is a new reading comprehension dataset, comprising of triples of passage-question-answer. The SQuAD dataset has more than 100,000 questions posed by crowd workers on a set of Wikipedia articles, wherein the answer to each question is a segment of text from the corresponding reading passage. In our research, the specific process mainly includes five steps:

1. Read the respective building regulations. The content of each regulation document is recorded during the reading process.
2. Process the building regulations. The regulation clause includes structured and non-textual data. Clauses are identified as units using punctuations as boundaries. Specific text contents in different formats (e.g., formula and chart), which cannot be simply used as data, are implemented as manually curated rules (see Section 3.1). All the section titles are merged with any sentence within the corresponding section to retain semantic information.
3. Construct passages. Treat each clause as a passage.
4. Ask questions. Questions based on the passage are asked. Each passage is asked one or more questions, and the same answer is enquired in different ways. At the same time, the answer in the passages is labeled.
5. Construct answers. The shortest consecutive string in the clause is constructed as the answer to the corresponding question.

The construction of dataset is the premise of the implementation of our methods. To this end, five civil engineers (two field engineers and three quality inspectors, all with an average of eight years of experience)

are invited to help create the dataset. The method and given examples are introduced for training. The output of each engineer is checked by the researchers. To verify the validity of the dataset, the Knuth-Morris-Pratt (KMP) algorithm [51], is used to match the answers and passages in the triples to check if an answer is completely contained in a passage. If an answer cannot be found in a passage, the triple is marked, and then reconstructed manually. In addition, the KMP is used to remove the repeated triples in the dataset. An example from the building regulation triples is shown in Table 4.

4.3. Model training and testing

The optimal BERT-based question answering model is fine-tuned through many rounds of model training with interspersed model validation. The implementation procedure and concrete methodologies involved in this workflow are illustrated as follows.

A total of 3500 triples were extracted from 19 regulations depicted in Section 4.1, which make up the dataset used to implement the question answering methodology. Cross-use validation sets are necessary for verification to support parameter optimization. Furthermore, if the model is trained and validated using all the collected data, then the situation that its ‘good’ performance could be limited to the previous data with the poor performance observed in new data can’t be detected [52]. Thus, some data should be reserved as testing sets to evaluate the model performance objectively. The dataset division for the collected data (the triple of construction quality inspection regulation) is conducted considering the aforementioned issues.

The collected texts are divided into the following three datasets: (1) training set, which is for training the model to be optimized; (2) validation set, which is for supporting the selection of the optimized model parameters; and (3) testing, which is for testing the performance of the established model. The three datasets are arranged in the following order of ratio: (1) training set; (2) validation set; and (3) testing set = 5:1:1. Specifically, the number of each set is as follows: (1) training (2500); (2) validation (500); and (3) testing (500). In addition, each set is equally divided into two parts, namely, definition and factoid, to help the model adapt to different types of clauses and test their answer. Furthermore, different types of text are added in reverse to ensure the effectiveness of the system according to the training results.

4.3.1. Model training

As previously mentioned, the pretrained BERT model on the Chinese Wikipedia is further trained and fine-tuned on the building regulation training set. The model optimization process (model training with interspersed validations) is used to obtain an optimized BERT-based QA model. Specifically, interspersed validations are performed with model training. This approach refers to the selection of optimized parameters according to the validation results. The data applied in the model

optimization are obtained from the training and validation sets. The optimized parameters are then verified by the testing set.

The developed BERT-based QA approach was performed on a server with an Intel Core Processor (Skylake) CPU, Tesla P4 with 8 GB video memory GPU, and 32 GB RAM. The Python programming language was also used. The TensorFlow 1.8.0 deep learning framework was also used as a basis to implement such approach. The developed model is trained with a batch size of four sequences with 256 tokens in each sequence for 1000 steps. The learning rate algorithm 3×10^{-5} of the Adams method, $\beta_1 = 0.9$, $\beta_2 = 0.999$, L_2 weight decay of 0.01, learning rate warm-up over the first 100 steps, and linear decay of the learning rate are used. A dropout probability of 0.1 is adopted in each layer [34].

4.3.2. Model testing and result

Three metrics are utilized to the performance of the question answering model: (1) F1 score; (2) exact match (EM); and (3) AVERAGE. The mean average precision, mean average recall, and mean average F-measure metrics, which are computed by averaging precision (P), recall (R), and F1 score over the questions, are used to evaluate the answers of questions. These measures are computed as follows:

$$P = \frac{|C|}{|A|}, R = \frac{|C|}{|Q|}, F1 = \frac{2PR}{P + R} \quad (7)$$

where $|C|$ is the number of correctly answered questions, $|A|$ is the number of extracted answers given by the model, and $|Q|$ is the number of all questions.

The answers in the QA model, such as “Beijing” versus “Beijing city” are the same. Two methods to count answered questions correctly are used:

1. Strict, where the extracted answer is the same with the answer given in the triple dataset; and
2. Fuzzy, where the extracted answer could be a synonym of the golden answer or an incomplete answer in form but complete in terms of necessary semantics.

Consequently, EM scores are used for testing. These scores measure the percentage of the prediction that matches one of the ground-truths answers exactly. The measure is computed as follows:

$$EM = \frac{C}{T}, \quad (8)$$

where C is the number of the answer returned, which is the same as the answer in the test set, and T is the number of the Total test question. Finally, AVERAGE scores are used to evaluate the performance of the QA model to consider the effects of F1 and EM. The AVERAGE is determined by:

$$AVERAGE = \frac{F1 + EM}{2} \quad (9)$$

The testing set is reserved to test the optimal model. The testing results of the QA model are illustrated in Table 5, in which F1, EM, and AVERAGE are recorded as the indexes of the model’s evaluation.

Comparing the AVERAGE scores of two types of clauses, factoid and definition types have relatively high scores of more than 80%, which proves a favorable application effect. Simultaneously, the score of the factoid type is close to 100%, while that of the definition type of the question is around 90%.

Table 4

Example from the building regulation triples.

Passage	The building quality shall be inspected and accepted according to the following requirements: (1) The acceptance of the project quality shall be carried out on the basis of the unit’s (i.e., contractor) own inspection and assessment. (2) The personnel of all parties participating in the project quality acceptance shall have the prescribed qualifications. (3) The quality of the inspection lot shall be accepted according to the master control and general projects. (4) Test blocks, test pieces and related materials related to structural safety shall be witnessed and tested according to regulations. (5) Before the concealment of works, the unit shall inform the supervision team to conduct an inspection, and create an acceptance document. Construction can only continue once the work has been accepted. (6) Sampling inspection shall be conducted carried out in accordance with the clauses before acceptance of important sub-projects involving structural safety, energy conservation, environmental protection, and usage functions.
Question	How to inspect and accept the quality of the inspection lot?
Answer	According to the master control projects and general projects

Table 5

F1, EM, and AVERAGE of the testing results.

Metrics	Factoid	Definition
F1 (%)	98.704	89.950
EM (%)	95.000	90.000
AVERAGE (%)	96.852	89.975

The performance (EM = 87.4, F1 = 93.2) of BERT applied to SQuAD outperforms the human performance (EM = 86.831 and F1 = 89.452) [34]. The performance of applying BERT in the building regulation field is slightly better. The results demonstrate that the BERT-based model for the regulation QA can be effectively utilized without incorporating sources of domain knowledge (knowledge patterns or templates) and heavy human feature engineering. The differences of the result between the factoid and definition shows the model is more accurate in answering the short text, since the longer the answer length, the more likely the model is to miss information in the answer extraction process.

Simultaneously, the dataset created for this research can be used in the future for similar regulation NLP tasks. The embeddings can be learned once and used thereafter for different tasks. Therefore, the workload associated with building, annotating, utilizing, and maintaining sources of domain knowledge can be avoided.

5. Prototype development and evaluation

Based on the described methodology, a prototype system for QA of building regulations is now developed and evaluated.

5.1. System implementation

The passage retrieval is implemented on the Lucene search engine (a popular open-source search engine built on TF-IDF). The related research experiment showed that the BERT model is unable to take advantage of additional answer passages that appear in the candidate passage pool [53]. The performance of F1 and EM usually stop increasing as the number of the candidate passages increases. To establish an ideal number of candidate passages, different values (i.e. 5, 10, 15, 20) are tested, and $D = 5$ is adopted. This can be expected since the number of relevant clauses in building regulations text is usually small.

A chatbot is deployed on the basis of the aforementioned framework. A QA system for the building regulations, referred as QAS4CQAR, in which users can interact with on WeChat's intelligent platform, enables users to engage with it and obtain answers quickly. A screenshot of the WeChat is shown in Fig. 3. The screenshot illustrates the diversity of questions that can be answered. One important consideration is the latency of the answer responses. A total of 100 questions are randomly

selected from the testing set to quantify the processing time, and average latencies are then recorded (measurements are taken on a machine depicted in the preceding section). An answer processing (on the CPU) averages 0.5 s per question.

Fig. 4 displays the management interface of the QAS4CQAR prototype. The dataset (including the annotation) can be created using the interface. Furthermore, the dataset can be managed (including add/delete/modify the dataset, initialize the parameters of the system) and train the model on the newly created dataset.

5.2. Evaluation

The processing of evaluating of the QAS4CQAR system involves its comparison with two frequently used methods: (1) the commercial Baidu search engine (a popular Chinese search engine, www.baidu.com); and (2) Regulation document database. In China, building practitioners are accustomed to using the Baidu search engine when encountering questions on regulations. Using internet search engines such as Baidu poses challenges in information content relatedness and quality. If Baidu is used then extensive links to documents rather than self-contained answers to specific questions are presented. Simultaneously, some practitioners may search for clauses in specific digital regulation document, which can be a time-consuming process.

As described in Zhong *et al.* [3] “the quality inspection involves many regulations, however, due to the reasons, such as unfamiliarity with or even lack of the regulation knowledge, or being overwhelmed of the amount of regulation text, from which the applicable regulation provisions should be referenced (this is a common case especially to the new hand)” (p. 1). An engineering practitioner (e.g., field engineer and quality inspector), for example, with limited working experience will no doubt need to retrieve information from the building regulations than those who have longer working experience and have more regulation knowledge. It is, however, outside the scope of this paper to compare and contrast the experience and knowledge of practitioners of the building regulations.

Ten experts (three field engineers and quality inspectors with average of 10 years of experience, two with average of six years of experience, and five with average of 2.5 years of experience) are asked to conduct a manual evaluation of the system. Two hundred questions are randomly selected from the testing set to evaluate QAS4CQAR



Fig. 3. A screenshot of the chatbot interface of QAS4CQAR.



Fig. 4. A screenshot of QAS4CQAR.

system. The Baidu search engine and regulation document database are also evaluated on the same set of questions for comparison. Such a comparison aims to determine the effectiveness of each of the three ways to answer the questions using a five-point Likert scale, which is divided to five categories: “strongly agree”, “agree”, “neutral”, “disagree” and “strongly disagree”, denoted as 9, 7, 5, 3 and 1 respectively. The questions focused on the following areas:

1. Ease of Use
2. Quality of Answers
3. Time Spent
4. The Overall Performance

Each expert is asked to test the Regulation document database (RDD) first, and Baidu secondly, and then QAS4CQAR finally. Each expert is presented with a mutually exclusive set of 20 out of 200 randomly selected questions. Experts provided overall score of one, three, five, seven, and nine in response to the above questions after the 20 questions tested. The score and average score with one decimal place are presented in Table 6.

These results demonstrate that the QAS4CQAR’ Ease of Use score 8.6 was very competitive with the other two ways. This result suggests that the WeChat chatbot interface is effective and beneficial in terms of its user-friendliness. Considering the quality of answers, the Regulation document database attained the best score of 8.8. This result is expected

as this database contains a rich domain-specific knowledge resource. Simultaneously, the QAS4CQAR system obtained the second-highest score in the quality of answers as they were segmentally extracted from the clauses of regulations. Justifying the exact answers (i.e., the sentence or the passages where the answers are extracted) is useful. The types of questions that belong to closed questions (factoid and definition) can yield explicit answers.

In the QAS4CQAR system, the average time spent on a question is less than 0.5 s, and the score of 8.8 was attained. By contrast, the digital regulation attained the worst score, to find the exact answers from the digital regulations, the average time is more than 30 s, an additional time, especially for the new hand, is occasionally needed to obtain the answer from the paper documents due to the burden on skimming and screening the recommended retrievals. In sum the experts suggested that the key benefits of the QAS4CQAR system were its ability to simplify the process to search building regulations and facilitate the process of learning about their use and application.

5.2.1. Limitations

Despite the success of the developed methodology in building regulation QA, the following limitations must be acknowledged. Similar to any other ML-based algorithms, the performance of the proposed method could vary, depending on the size of the training and testing datasets. The methodology is only evaluated on the given dataset. Thus, the sample size can be improved in the future to further verify and refine

Table 6
Evaluation score of QAS4CQAR, Baidu, and RDD.

Method	Metric	Score										Average score
QAS4CQAR	Ease of Use	9	9	9	7	9	9	9	7	9	9	8.6
	Quality of Answers	7	7	7	7	7	7	9	7	7	7	7.2
	Time Spent	9	9	9	7	9	9	9	9	9	9	8.8
	Overall Performance	9	9	9	7	7	9	9	7	9	7	8.2
Baidu	Ease of Use	5	5	5	5	7	7	7	7	7	5	6.0
	Quality of Answers	5	5	5	5	3	5	7	5	5	5	5.0
	Time Spent	7	7	5	7	5	5	7	7	7	7	6.4
	Overall Performance	5	5	5	5	5	5	7	7	7	5	5.6
RDD	Ease of Use	7	7	7	5	7	7	5	5	5	5	6.0
	Quality of Answers	9	9	9	9	9	9	7	9	9	9	8.8
	Time Spent	7	7	7	7	5	7	5	5	5	5	6.0
	Overall Performance	7	7	7	7	7	7	5	5	7	5	6.4

the performance of the presented approach.

The pretrained BERT model from public database (Chinese Wikipedia) may lack domain knowledge, which has an effect on being used for the building regulations extraction. In addition, only the natural language clauses are covered in the model, other types of clauses such as tables in regulations need transformed into text, which might lead to missing semantic. Meanwhile, for the evaluation of the prototype system, only the expert's comparison evaluation is conducted between three methods of QAS4CQAR system, Baidu, and Regulation document database.

In this paper, a VSM formed the basis of the retrieval module without considering semantic features. The ontology-based information retrieval approach has several benefits: (1) handling synonyms and morphological changes; and (2) allowing semantic query expansion and reasoning through inclusion and domain relations. In the future, we suggest that ontology and semantic technology will be able to accommodate a retrieval module.

6. Conclusion

An end-to-end methodology for the QA of building regulations by integrating information retrieval based on a deep learning model of an improved BERT is presented and used to create the QAS4CQAR system. The methodology and QAS4CQAR system are evaluated and compared with the Baidu search engine and digital regulation documents. The experiments demonstrate that the QAS4CQAR system provides practical benefits for practitioners.

The paper's contribution is twofold. A methodology that combines NLP and deep learning to be able to address queries raised about the building regulations is proposed. The performance of the methodology is evaluated. In contrast to traditional template-based question answering methodology focusing on the specific patterns of text document and small corpus with low scalability, as an end-to-end learning approach, the proposed methodology has powerful feature representation and learning capability in addressing building regulations. A chatbot of question answering system for the building regulation, hereafter called QAS4CQAR, is developed to provide exact answers to questions. This system is experimentally evaluated on the given dataset. The expert review is conducted for the feasibility and the potential of the system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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